

A price tracker that tells you **what changed**

A watcher that turns a product catalogue into a spreadsheet, keeps every run, and leads its report with the changes.

THE PROBLEM

A daily export of every product is a file nobody opens. What a buyer needs is an alert when a price drops, an item goes out of stock, or something new appears.

WHAT I BUILT

- Crawler with two engines, fast HTTP and headless Chromium, that return identical rows
- Every run stored in SQLite with a CSV export
- Change detection for price drops, price rises, stock flips, new and removed products
- Single file HTML report in dark and light with an empty state
- Polite delay and an identifying user agent on books.toscrape.com, a store made for scraping practice

MEASURED RESULTS

32 products

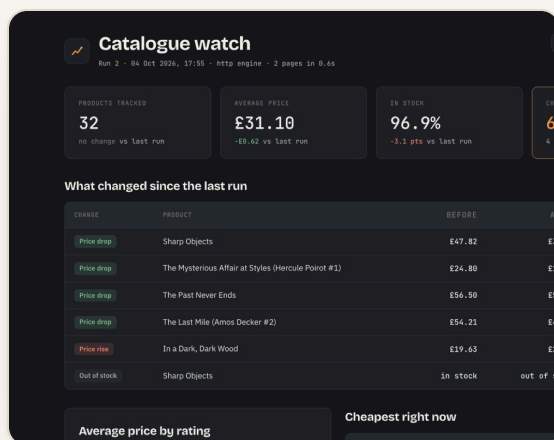
across 2 pages in about 1 second with the HTTP engine

6 → 0

changes found on run 2, none on the unchanged run 3

9 tests

parsing, pagination, each change type, the report and the CSV



Change report for run 2

run-2.csv

	A
1	title
2	1st to Die (Women's Murder Club #1)
3	A Murder in Time
4	A Study in Scarlet (Sherlock Holmes #1)
5	A Time of Torment (Charlie Parker #14)
6	Blood Defense (Samantha Brinkman #1)
7	Boar Island (Anna Pigeon #19)
8	Career of Evil (Cormoran Strike #3)
9	Delivering the Truth (Quaker Midwife Mystery...)
10	Extreme Prey (Lucas Davenport #26)
11	Hide Away (Eve Duncan #20)
12	In a Dark, Dark Wood

CSV export as a spreadsheet

```
$ python -m watcher.cli run --max-pages 2
Catalogue watcher (sample project)
source books.toscrape.com, Mystery (day
engine http
database data/catalog.db

page 1 20 products index.html
page 2 12 products page-2.html

32 products stored as run 2 in 0.6s
6 changes against run 1, including 4 price
csv data/run-2.csv
report data/report-2.html
```

A real run in the terminal

The practice store never changes prices, so run 2 crawled a local copy with four prices and one stock line edited. The diff logic and both crawls are real.

STACK

Python 3 httpx BeautifulSoup

Playwright SQLite Jinja2

Code github.com/vanthienha199/catalog-watcher-demo